

ARTIFICIAL INTELLIGENCE

Student Class Notes

Chapter 4: Intelligent Agents and Environments

Chapter 5: Nature of the Environment

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INTELLIGENT AGENTS AND ENVIRONMENTS

4.1 Learning Outcomes

- Define an intelligent agent and its environment.
- Distinguish sensors, actuators, percepts, and actions.
- Explain percept sequences and the agent function.
- Describe rational behaviour and performance measures.
- Use the PEAS framework to specify task environments.
- Identify agents in familiar real-life systems.

4.2 Introduction

Artificial Intelligence can be viewed as the study of intelligent agents. An agent receives information from the world around it, processes that information, selects an action, and then affects the world through that action. This perception-action cycle is central to robotics, software assistants, autonomous vehicles, recommender systems, medical decision support, and many other AI applications.

Core idea: An agent perceives its environment through sensors and acts upon that environment through actuators.

ENVIRONMENT	SENSORS	AGENT	ACTUATORS	ENVIRONMENT
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Information flows from the environment to the agent as percepts. Actions flow from the agent back to the environment.

4.3 Agent, Environment, Sensors and Actuators

4.3.1 Agent

An agent is any entity that can perceive its environment and act upon it. An agent may be biological (a human), mechanical (a robot), or purely software-based (a chatbot or search assistant).

4.3.2 Environment

The environment is the external world in which the agent operates. It supplies information to the agent and is affected by the agent's actions. For a chess program, the environment is the chessboard and opponent. For a self-driving car, it includes roads, traffic, pedestrians, signals, passengers, weather, and other vehicles.

4.3.3 Sensors

Sensors are the means by which an agent receives information. Human sensors include eyes, ears, skin, nose, and tongue. Robot sensors may include cameras, GPS, microphones, temperature sensors, radar, and lidar. A software agent may sense files, clicks, database records, messages, or network packets.

4.3.4 Actuators

Actuators are the means by which an agent performs actions. Human actuators include hands, legs, and voice. A robot may use wheels, motors, grippers, brakes, or a speaker. A software agent may display text, send a message, update a file, or call another software service.

Agent	Environment	Sensors	Actuators	Goal
Human	Room / society	Eyes, ears, skin	Hands, legs, voice	Complete a task safely
Robot vacuum	Room floor	Dirt sensor, camera	Wheels, suction motor	Clean efficiently
Self-driving car	Road and traffic	Camera, GPS, radar/lidar	Steering, brake, accelerator	Reach destination safely
Software assistant	Computer / network	Files, user input, messages	Display, write, send	Help the user
Thermostat	Room	Temperature sensor	Heater/cooler switch	Maintain temperature

4.4 Percepts, Percept Sequence, Action and Agent Function

4.4.1 Percept

A percept is the information received by an agent at a particular moment. For example, “traffic signal is red” is a percept received by a driver or autonomous vehicle.

4.4.2 Percept Sequence

A percept sequence is the complete history of percepts received so far. A rational decision may depend not only on the current input but also on previous observations.

4.4.3 Action

An action is what the agent does in response to its percepts. Examples include moving left, applying brakes, displaying a route, making a recommendation, or sending an alert.

4.4.4 Agent Function

The agent function maps every possible percept sequence to an action. In compact mathematical form:

$$f : P^* \rightarrow A$$

Here, P^* denotes the set of all possible percept sequences and A denotes the set of available actions. The agent program is the practical implementation of this mapping on a particular architecture.

4.4.5 Vacuum-Cleaner World Example

Consider two locations, A and B. The vacuum agent senses its current location and whether that location is dirty. Its actions are Left, Right, Suck, and No-Op.

Percept	Action
A, Dirty	Suck
A, Clean	Move Right
B, Dirty	Suck
B, Clean	Move Left

This simple example shows how an agent converts percepts into actions. If the dirt sensor fails, the same program may behave poorly because its percepts are incomplete or incorrect.

if Status == "Dirty": return Suck elif Location == "A" (Left Room): return Right elif Location == "B" (Right Room): return Left else: return No-Op

4.5 Rational Agents and Performance Measures

4.5.1 Rational Agent

A rational agent chooses the action that is expected to maximize its performance measure, given the percept sequence, prior knowledge, available actions, and current uncertainty. Rationality does not mean perfection. A rational agent can still make a poor decision if its information is incomplete or its model is inaccurate.

Rationality means: the best possible action using the information and resources currently available.

4.5.2 Performance Measure

A performance measure defines what counts as success. Good agent design begins by specifying the desired outcome rather than only describing the actions.

Agent	Possible Performance Measures
Google Maps	Safe arrival, low travel time, legal route, low cost
Robot vacuum	Cleanliness, low energy use, low noise, short cleaning time
Medical assistant	Diagnostic accuracy, safety, explainability, low false alarms
Chess agent	Win, avoid illegal moves, use time efficiently
Self-driving taxi	Safety, comfort, correct destination, legal driving, low fuel use

4.6 PEAS Framework

PEAS is a systematic framework for describing an agent's task environment before designing the agent.

P - Performance Measure

E - Environment

A - Actuators

S - Sensors

4.6.1 PEAS Example: Self-Driving Taxi

PEAS Element	Description
Performance Measure	Safety, correct destination, minimum time/cost, passenger comfort, legal driving
Environment	Roads, traffic, pedestrians, signs, weather, passengers, other vehicles
Actuators	Steering, accelerator, brake, horn, indicators, display
Sensors	Cameras, GPS, speedometer, radar/lidar, microphones, proximity sensors

4.6.2 Additional PEAS Examples

Agent	PEAS Summary
Smart Attendance System	Correct identification; classroom; camera/display; camera, timetable, student database
Medical Diagnosis Assistant	Accurate and safe recommendation; patient/clinic; report/display; symptoms, tests, records
Chess Agent	Win legally and efficiently; board/opponent; move output; current board position
Food-Delivery Robot	Safe and timely delivery; campus/road; wheels, arm, display; camera, GPS, proximity sensors

4.7 Solved Examples

Example 1: Google Maps

Agent: route-planning software. Environment: road network and traffic. Sensors: GPS, map database, traffic reports, user input. Actuators: visual route, voice instructions, alerts. Performance: safe and efficient arrival. A rational route is not always the geometrically shortest route; traffic, closures, tolls, and safety also matter.

Example 2: Smartwatch Health Monitor

Agent: health-monitoring software. Environment: the user and surrounding activity. Sensors: heart-rate sensor, accelerometer, oxygen sensor, GPS. Actuators: vibration, display, warning, data synchronization. Performance: useful alerts with few false alarms and respect for privacy.

Example 3: Chatbot

Agent: conversational software. Environment: the user conversation and available tools/data. Sensors: user messages, uploaded files, tool outputs. Actuators: text, images, generated files, or tool calls. Performance: relevance, correctness, safety, clarity, and helpfulness.

4.8 Chapter Summary

- An agent perceives an environment through sensors and acts through actuators.
- A percept is a single observation; a percept sequence is the complete history of observations.
- The agent function maps percept sequences to actions.
- A rational agent chooses the action expected to maximize performance.
- PEAS specifies Performance Measure, Environment, Actuators, and Sensors.
- Good agent design starts with a clear task environment and performance measure.

4.9 Practice Questions

A. Multiple-Choice Questions

1. An agent receives information through:
- A. Sensors B. Actuators C. Goals D. Rewards

2. The means by which an agent acts are called:
A. Percepts B. Sensors C. Actuators D. States
3. A percept sequence is:
A. One action B. History of all percepts C. Only the current state D. A performance score
4. The agent function maps:
A. Actions to sensors B. Percept sequences to actions C. Goals to data D. Environments to programs
5. In PEAS, P stands for:
A. Program B. Percept C. Performance Measure D. Probability
6. A rational agent aims to:
A. Be perfect B. Maximize expected performance C. Ignore uncertainty D. Use no sensors
7. Which is an actuator of a self-driving car?
A. GPS B. Camera C. Brake D. Radar
8. Which is a sensor of a robot vacuum?
A. Suction motor B. Wheel C. Dirt detector D. Brush
9. The external world in which an agent operates is the:
A. Architecture B. Environment C. Program D. Function
10. Agent = Architecture + _____.
A. Environment B. Program C. Sensor D. Percept

Answer Key: 1-A, 2-C, 3-B, 4-B, 5-C, 6-B, 7-C, 8-C, 9-B, 10-B

B. Short-Answer Questions

1. Define an intelligent agent.
2. Differentiate sensors and actuators.
3. What is a percept sequence?
4. Explain rationality in AI.
5. What is a performance measure?
6. Expand PEAS.
7. Write the PEAS description of a chess agent.
8. Distinguish an agent function from an agent program.

C. Long-Answer Questions

9. Explain agents and environments with a labelled diagram and examples.
10. Describe a rational agent and the factors affecting rationality.
11. Explain the PEAS framework for an automated taxi.
12. Using a suitable example, explain percepts, percept sequences, actions, and the agent function.

D. Assignment

Choose one intelligent system - Google Maps, a smartwatch, chatbot, robot vacuum, or self-driving car. Prepare a PEAS table and identify at least four properties of its environment.

NATURE OF THE ENVIRONMENT

5.1 Learning Outcomes

- Explain why environment properties influence agent design.
- Differentiate fully and partially observable environments.
- Distinguish deterministic and stochastic environments.
- Classify episodic/sequential, static/dynamic, discrete/continuous, single/multi-agent, and known/unknown environments.
- Classify familiar AI tasks such as chess, driving, diagnosis, and image recognition.

5.2 Introduction

An intelligent agent cannot be designed properly without understanding the environment in which it will operate. A chess program, medical assistant, robot vacuum, and self-driving car face very different kinds of uncertainty, change, interaction, and information. Therefore, AI environments are classified along several dimensions.

Design principle: The more uncertain, dynamic, continuous, and multi-agent an environment is, the more sophisticated the agent must be.

5.3 Dimensions of Task Environments

5.3.1 Fully Observable vs Partially Observable

A fully observable environment provides the agent with all information relevant to choosing an action. A partially observable environment hides some information or provides noisy/incomplete sensor data.

Type	Meaning	Examples
Fully observable	All relevant state information is available	Chess board, Sudoku
Partially observable	Some state information is hidden or uncertain	Driving in fog, medical diagnosis

5.3.2 Deterministic vs Stochastic

In a deterministic environment, the next state is completely determined by the current state and the chosen action. In a stochastic environment, outcomes involve uncertainty or probability.

- Deterministic example: a well-defined puzzle move.
- Stochastic example: driving, where other road users may behave unpredictably.

5.3.3 Episodic vs Sequential

In an episodic environment, each decision is independent of previous decisions. In a sequential environment, current actions affect future states and later choices.

- Episodic example: classifying independent images.
- Sequential example: chess, route planning, and medical treatment.

5.3.4 Static vs Dynamic

A static environment does not change while the agent is deciding. A dynamic environment may change during the decision process. A semidynamic environment has a stable world state, but the performance measure changes with time, as in timed chess.

- Static example: crossword puzzle.
- Dynamic example: road traffic.
- Semidynamic example: chess with a clock.

5.3.5 Discrete vs Continuous

A discrete environment has a countable number of states, percepts, or actions. A continuous environment contains values that can vary across a range, such as speed, time, temperature, position, or steering angle.

- Discrete example: chess moves.
- Continuous example: controlling a car's steering and speed.

5.3.6 Single-Agent vs Multi-Agent

A single-agent environment contains one decision-making agent whose success does not depend on strategic interaction with other agents. A multi-agent environment contains other agents whose actions influence outcomes. Multi-agent settings may be cooperative, competitive, or mixed.

- Single-agent example: solving Sudoku.
- Multi-agent example: chess, football, traffic, and online marketplaces.

5.3.7 Known vs Unknown

In a known environment, the agent knows the rules governing actions and outcomes. In an unknown environment, the agent must learn how the environment behaves. Known/unknown is different from observable/partially observable: one concerns knowledge of rules, while the other concerns access to the current state.

- Known example: chess rules are fully specified.
- Unknown example: a robot exploring an unfamiliar planet or building.

5.4 Comprehensive Comparison Table

Dimension	Type 1	Type 2	Typical Contrast
Observability	Fully observable	Partially observable	Chess vs driving in fog
Outcome	Deterministic	Stochastic	Puzzle vs traffic
Dependence	Episodic	Sequential	Image classification vs chess
Change over time	Static	Dynamic	Crossword vs road traffic
State/action space	Discrete	Continuous	Chess move vs steering angle
Number of agents	Single-agent	Multi-agent	Sudoku vs football
Knowledge of rules	Known	Unknown	Chess vs exploration

5.5 Complete Case Classifications

5.5.1 Chess

- Fully observable
- Deterministic
- Sequential
- Static or semidynamic
- Discrete
- Multi-agent
- Known

Both players see the board and know the rules. Each move affects future moves. If a clock is used, time affects performance, making the task semidynamic.

5.5.2 Self-Driving Car

- Partially observable
- Stochastic
- Sequential
- Dynamic
- Continuous
- Multi-agent
- Known and partly learned

The car cannot know every intention or hidden obstacle. Traffic changes continuously, actions affect later states, and speed/position vary continuously. The system must interact with drivers, cyclists, and pedestrians.

5.5.3 Medical Diagnosis Assistant

- Partially observable
- Stochastic
- Sequential
- Dynamic
- Mixed discrete/continuous
- Multi-agent
- Partly known

The patient's complete internal state is hidden. Tests may be uncertain, treatments affect future outcomes, and the assistant interacts with doctors and patients.

5.5.4 Image Classification

- Usually fully observable at the image level
- Often treated as deterministic
- Episodic
- Static
- Mostly continuous input with discrete labels
- Single-agent

- Known task definition

Each image can normally be classified independently. Pixel values are continuous or high-dimensional, while output labels are discrete.

5.5.5 Robot Vacuum

- Partially observable
- Often stochastic
- Sequential
- Dynamic
- Mixed discrete/continuous
- Usually single-agent
- Known or partly known

The robot may not see the entire room. Movement and sensing can be noisy, earlier cleaning choices affect later efficiency, and people may change the environment.

5.6 Why Environment Classification Matters

Environment classification is not only theoretical. It guides the selection of sensors, memory, planning methods, probabilistic models, learning algorithms, and safety mechanisms.

Environment Property	Design Consequence
Partially observable	Use memory, state estimation, belief states, or more sensors
Stochastic	Use probability, risk analysis, and robust decisions
Sequential	Use planning and consider long-term consequences
Dynamic	Respond quickly and update continuously
Continuous	Use numerical models and continuous control
Multi-agent	Model cooperation, competition, communication, or game strategy
Unknown	Use exploration and learning

5.7 Chapter Summary

- Fully observable means all relevant state information is available; partially observable means some information is hidden or noisy.
- Deterministic environments have predictable transitions; stochastic environments involve uncertainty.
- Episodic tasks contain independent decisions; sequential tasks have long-term consequences.
- Static environments do not change during deliberation; dynamic environments do.
- Discrete environments have countable states/actions; continuous environments contain ranges of values.
- Single-agent tasks involve one decision maker; multi-agent tasks involve strategic or cooperative interaction.
- Known environments have known rules; unknown environments require learning.

5.8 Practice Questions

A. Multiple-Choice Questions

1. Chess is generally considered:
A. Partially observable B. Fully observable C. Unknown D. Episodic
2. Road traffic is typically:
A. Static B. Dynamic C. Episodic D. Single-agent
3. Image classification is usually:
A. Sequential B. Episodic C. Multi-agent D. Unknown
4. A deterministic environment has:
A. Predictable transitions B. No sensors C. Many agents D. No actions
5. Steering angle is usually:
A. Discrete B. Continuous C. Unknown D. Episodic
6. Football is a:
A. Single-agent task B. Multi-agent task C. Static task D. Deterministic task
7. Medical diagnosis is often:
A. Fully observable B. Partially observable C. Static only D. Deterministic only
8. Chess with a clock is best described as:
A. Static B. Semidynamic C. Unknown D. Continuous
9. In an unknown environment, an agent must:
A. Know all rules already B. Learn how the environment behaves C. Avoid actions D. Use no memory
10. Driving is generally:
A. Episodic B. Sequential C. Static D. Single-agent

Answer Key: 1-B, 2-B, 3-B, 4-A, 5-B, 6-B, 7-B, 8-B, 9-B, 10-B

B. Short-Answer Questions

13. Differentiate fully and partially observable environments.
14. What is a stochastic environment?
15. Differentiate episodic and sequential tasks.
16. Explain static, dynamic, and semidynamic environments.
17. Differentiate discrete and continuous environments.
18. What is a multi-agent environment?
19. Differentiate known and unknown environments.
20. Why is a self-driving environment partially observable?

C. Long-Answer Questions

21. Explain the major dimensions used to classify AI task environments, with examples.
22. Classify the environment of a self-driving car and justify each property.
23. Compare the environments of chess, medical diagnosis, and image classification.

24. Explain how environment properties influence agent design.

D. Classification Exercise

Classify each of the following environments using all seven dimensions: online shopping recommendation, autonomous drone delivery, weather forecasting, smart classroom attendance, and robotic surgery.

References and Further Reading

- 25. Russell, S. J., and Norvig, P. Artificial Intelligence: A Modern Approach, 4th ed., Chapter 2: Intelligent Agents.
- 26. Artificial Intelligence: A Modern Approach official resources, University of California, Berkeley:
<https://aima.cs.berkeley.edu/>
- 27. Stanford Encyclopedia of Philosophy, “Artificial Intelligence,” section on intelligent agents.
- 28. Faculty scripts supplied for this course: “Agents and Environments” and “Nature of the Environment.”